

DAD-MCNN: DDoS Attack Detection via Multi-channel CNN

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ABSTRACT

With the continuous development of web services, the web security becomes more and more important. Distributed Denial of Service (DDoS) attack as one of the most common form of attacks, has produced serious economic damages. DDoS attack detection as one of main defense methods is suffered increasing attention by researchers. Most of them use machine learning methods to make good detection performance. However, there are still gaps between real detection rate and expected one, conventional machine learning methods are limited compared with deep learning. In this paper, we propose DAD-MCNN, a multi-channel CNN(MC-CNN) based DDoS attack detection framework, which can fully utilize information from a huge amount of network packages and set up an earlier warning system. Our contributions can be summarized as follows: (1) we propose a new preprocessing method for the network dataset. (2) MC-CNN is applied to detect DDoS attack and the detection result is decided by data in respective channels. (3) We use incremental training method to optimize training procedures and time in MC-CNN. (4) The experiment result shows that MC-CNN has the highest accuracy compared with conventional machine learning methods. The result also proves that our approach has performed well not only in DDoS attack detection but also in other anomaly attack detection.

CCS Concepts

• Networks→Network properties→Network security→Denial-of-service attacks

Keywords

DDoS attack detection; networking and information security; multi-channel; deep learning.

1. INTRODUCTION

With the globalization of the economy and the popularization of the e-commerce, the competitive pressure of modern enterprises is increasing. Web service is the most important implementation of service computing technology. It breaks the limits of interaction between heterogeneous platforms. Internet companies can meet the flexible demand of consumer and market by combining existing services.

Web service allows companies to avoid the high cost of

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redevelopment but also improves the application quality through service selecting. However, due to the universality and importance of web services, the security problem of web services is becoming more and more prominent and one of the most common ways to attack is DDoS (Distributed Denial of Service) attack. A DoS attack can be described as an attack designed to render a computer or network incapable of providing normal services. A DDoS attack uses many computers to launch a coordinated DoS attack against one or more targets [1]. When the interface of a service provider is attacked, the related service requester can also be affected on a large scale, which will cause the economic losses.

DDoS detection is one of the main defense methods. In recent years, various methods have been proposed. Zhou et al. [2] constructed a Real-time Frequency Vector (RFV) and real-timely characterized the traffic as a set of models to predict application-layer distributed denials of service attacks. Spyridopoulos et al. [3] proposed a game theoretic defense framework to against DoS/DDoS cyber attacks. We will discuss more machine learning methods for detection in Section 2. Most of these methods need to analyze the packet and establish the detection model according to the features of the instances after extracting the static features.

However, with the development of networks infrastructure and architecture, the information that networks contains is rising. Facing these voluminous network data, the conventional machine learning method becomes limited to making full use of huge information. Therefore, it is high time to establish a DDoS attack framework which can use mass information and have high accuracy. CNN is a deep model applied to not only image classification, but also many other fields [4]. It can extract features automatically and have higher classification performance compared with other methods. The MC input strengthens the connection between each feature. The practices in other areas show that MC-CNN architecture can effectively improve the robustness against errors [5].

Inspired by above methods, we propose a DDoS attack detection and warning system based on MC-CNN, denoted as DAD-MCNN. MC-CNN is our core work in the DAD-MCNN. Firstly, we split features based on the source of different levels like time level, space level, packet level and so on. After splitting, we have several sub-datasets. We put them into different channels respectively and train MC-CNN. The MC-CNN has input layer, convolutional layer, pooling layer and output layer. The number of channels is depending on the number of feature groups. When the loss converges, we get an excellent detection model with high accuracy.

Our contributions can be summarized as follows:

(1) We propose DAD-MCNN, a detection and early warning framework for DDoS attack. The detection system gives a result based on the information of real-time traffic and network packages.

(2) We use MC-CNN to detect DDoS attack. The number of channels is decided by the number of feature groups. We feed

sub-datasets into channels respectively. It allows CNN to utilize the information on all feature levels.

(3) We use incremental training method to training MC-CNN. At first, we train a MC-CNN with 1 convolutional layer in each channel. Then we add a new convolutional layer to MC-CNN and continue training. Repeat previous steps and finally we get the MC-CNN with 3 convolutional layers in high accuracy. This method optimizes training procedures and time compared with train target structure MC-CNN directly.

(4) In the experiments, we train DDoS attack detection model based on conventional machine learning methods, LSTM and our MC-CNN. The results show that MC-CNN has the highest accuracy compared with state-of-art methods.

The rest of this paper is organized as follows. Section 2 introduces DDoS attack detect method based on machine learning and CNN. In Section 3, we propose the framework. In Section 4, we describe the experimental process and analyze the experimental results. The summary and future work are presented in Section 5.

2. RELATED WORK

2.1 DDoS Attack Detection Based on Machine Learning

Lots of machine learning algorithms are applied to DDoS defense and achieved good performance. Xu et al. [6] proposed a method using Hidden Markov Models (HMM) and cooperative reinforcement learning to detect DDoS attack traffic by monitoring source IP addresses. The detectors are distributed among the mediate network nodes or near the sources of DDoS attacks and HMMs are used to establish a profile for normal traffic based on the frequencies of new IP addresses. Berral et al. [7] proposed a framework to detect abnormal behaviors. In the framework, nodes in an intermediate network share information about their local traffic observations, improving their global traffic perspective and each node use native Bayes algorithm learning information independently.

Neural network is also widely used in detection of DDoS attack after extracting information on each network packet. Manikopoulos et al. [8] used statistical models and neural network classifiers to detect anomalous network conditions. Hsieh et al. [9] proposed DDoS detection method based on Neural Networks, implemented in the Apache Spark cluster. Zhao et al. [10] implemented and validated a neural network based DDoS detection system using Hadoop and HBase. Yuan et al. [11] proposed a deep learning based DDoS attack detection approach. They used LSTM to make detection considering the correlation in time series. Their method gives detection for grouping packet and performs better comparing with conventional machine learning method.

2.2 Convolutional Neural Network

Convolutional Neural Network (CNN), inspired by the visual cortex of animals, is the widely used deep learning framework and is widely used for object recognition tasks. Due to its unique characteristics, CNN is applied to many other fields like text detection and recognition [12], scene labeling [13] and more. In the conventional classification tasks, the extracted features are often pre-designed features based on statistical regularities or prior knowledge. So, it cannot represent the instance information comprehensively and accurately. CNN puts forward the concept of receptive field and shares weights, which not only greatly

reduces the parameters of training, but also reduces the complexity of network model.

A typical CNN structure mainly consists of the input layer, the convolutional layer, the pooling layer, the full connect layer and the output layer [14]. The training of CNN mainly uses the forward propagation and BP (Back Propagation) algorithm to learn the layer-connection weights, bias and other parameters. It is a supervised learning framework that requires the data as input and the corresponding labels to optimize the network parameters.

3. METHODOLOGY

DAD-MCNN is a framework of DDoS attack detection and warning. Monitoring and collecting information from web service ports or network nodes are basic parts of our work. Before training MC-CNN, we should split instances features to utilize information on the different level of feature groups. After training MC-CNN, we set up a real-time detection system. The overall flowchart of the framework is shown in Fig. 1. We discuss the design and application of this framework in this section.

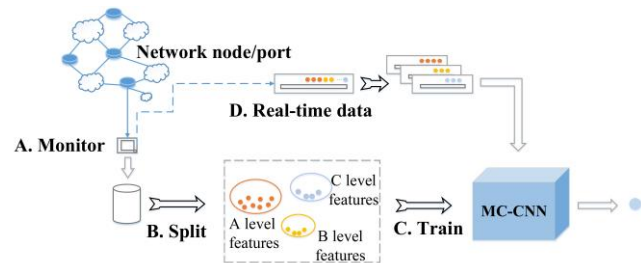


Figure 1. Flowchart of the DAD-MCNN. Data collected by monitor can be grouped into several levels. A, B, C levels in flowchart can be packet level, traffic level and so on. Dataset is split by feature groups into several parts and then put into independent channels to train MC-CNN.

3.1 Monitor and Collect Information

Monitor network traffic and packets are the basic and essential parts of the whole framework. During the running time, we keep monitoring network and analyzing the real-time indicators. Then we input data into the database. The indicators are duration, mean length of a flow, average size of the packet and more traffic features. Researchers can check historical information to find when the attack start and end manually. It not only provides training data but also sends real-time data to detection model after warning system set up.

3.2 Split Data Features

Different from detection issues in other fields. Both network traffic and DDoS attack are continuous processes. It suggests that records in the database contain information not only in network dimension but also in time dimension. At the same time, network information also can be divided into traffic part and packet part. Thus, making full use of features in every level is important. In the traditional way, researchers input instances to a single channel model to make the classification. However, in DAD-MCNN, as shown in Fig. 2, we split features into feature groups according to different levels, like traffic level, packet level, host level. The split standard is flexible according to the different views on features.

According to split standard, a dataset can be divided into several sub-datasets. The instances in each sub-dataset only contain a part

of the features, and the number of instances does not change after dividing.

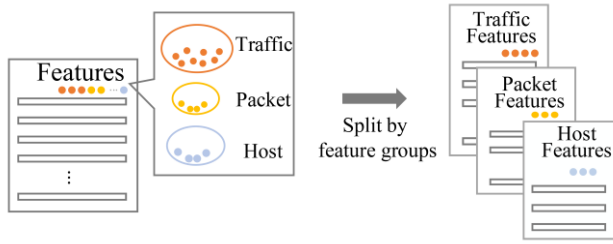


Figure 2. Split data features.

3.3 Train MC-CNN by Incremental Training Method

After splitting features and dividing dataset, we have several sub-datasets. Then, we build MC-CNN model. This neural network has an input layer, convolutional layers, pooling layers, full connect layers and an output layer. The construction of MC-CNN is shown in Fig. 3. Different from general CNN, we have several channels according to the number of feature groups. We feed different sub-datasets into the corresponding channels respectively. Data in each channel through convolution and pooling. The model is trained until the loss converges. We train MC-CNN in target structure by incremental training method. Firstly, we train a MC-CNN which only has one convolutional layer in each channel. Then, we remain parameters in trained layers and add a new convolutional layer to the MC-CNN model trained before and train it. Repeat previous steps and finally we get the MC-CNN with target structure.

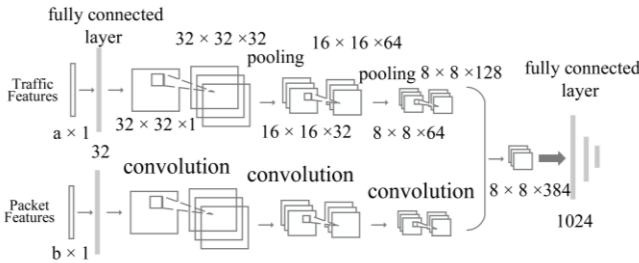


Figure 3. Structure of MC-CNN (2 channels). a, b is number of features in each sub-dataset.

3.4 Set Up Real Time Detection System

After previous steps, we have a trained MC-CNN model with high accuracy for DDoS attack detection. With this model and monitoring measures mentioned above, a real time detection system can be set up now. When records are collected by a monitor, they will be split by existing rules. Then, split records are put into corresponding channels. If DDoS attack exists, the model will give a warning.

4. EXPERIMENTS

We conduct a series of tests in KDDCUP99 [15], CICIDS2017 [16] and compare MC-CNN with LSTM and other swallow machine learning methods. The result shows that MC-CNN is more outstanding compared with other methods.

4.1 Dataset

We evaluate DAD-MCNN on CICIDS2017 dataset and KDDCUP99. Table 1 shows main information on two datasets.

CICIDS2017 dataset contains benign and the most up-to-date common attacks. It also includes the results of the network traffic analysis, timestamp, source and destination IPs, source and destination ports, protocols and labels. The data capturing periods started at 9:00 am, Monday, July 3, 2017 and ended at 17:00 on Friday, July 7, 2017, for a total of 5 days. We choose Wednesday, July 5, 2017 which contains DDoS attacks. The file size in this day is 13GB. There are 83 features in the dataset, containing both traffic information and packet information like IP, protocol, flow duration, etc. We do preprocess before the experiment. For source and destination IPs, we split them into four parts by dot and each part becomes a feature. We normalize the remaining numeric features. Finally, we have a dataset with 75 features and 1 label.

KDDCUP99 is one of the most commonly used datasets in the intrusion detection research. The types of attacks are shown in Table 2. The whole dataset consists of 4898431 records, each with 41 features such as duration, protocols, service, etc. In the experiment, we take the dataset in 2-class (normal, attack) and 5-class (Normal, Dos, Probe, R2L, U2R). We also transform string features into numeric type before training. Some special features on a large scale, we use $f = \log(x)$ to make their distribution uniform before training. It makes model to be trained better.

From Table 3, we find an imbalance problem in KDDCUP99. The instances of R2L, U2R are quite rare compared with Dos instances. So, we take random over-sampling for R2L and U2R to expand the size of instance. Considering our main detect target is Dos attack, we randomly delete quite number of Dos instance until the ratio between Dos and Normal is 1: 1. We do this adjustment process in each repeated experiment.

Table 1. Main information of datasets.

Dataset	#of features	All	#of instances anomaly (%)
KDDCUP99	41	489843	392565(80.14%)
CICIDS2017	83	225745	41835(18.53%)

Table 2. Feature class in KDDCUP99.

Feature class	Number
Basic features of individual TCP connections	9
Content features within a connection suggested by domain knowledge	13
Traffic features computed using a two-second time window	9
Traffic features based on host	10

Table 3. The original ratio of class in KDDCUP99.

Normal	Dos	R2L	U2R	Probe
19.69%	79.24%	0.22%	0.01%	0.83%

4.2 Experimental Settings

The process of our experiment follows the procedure in Fig. 1. Considering we use the dataset to train and test MC-CNN, we emphatically introduce step 2 and step 3. In the experiment, we use KDDCUP99 and CICIDS2017. We split the instances according to the features and input them to different channels

respectively. After training, we evaluate and compare MC-CNN accuracy with other methods.

Data format transformation

We find that features in CICIDS2017 can be split into 2 groups. One is based on package information and the other is based on traffic indicators. So, for CICIDS2017 dataset, we set two channels to train model. From Table 2, we find features in KDDCUP99 can be split into 4 groups. So, we use 4 channels in the training.

Train MC-CNN

We put split data into the corresponding channels respectively. In each channel, the data will through full connect layers and be increased dimension. Then it through convolution layer and pooling layer. Finally, matrices in each channel will be added up and through full connect layers. We use a global average pool to reduce parameter complexity of the full connection layer. The epoch in the train is 10. We also try to change size of training dataset to evaluate the efficiency of the classifier.

4.3 Experiment Results and Discussion

We use 3 layers of LSTM network, CNN, random forest (RF), SVM, C4.5 and KNN as our baselines. In the evaluation process, we use classification accuracy to show the performance compared with baselines. Table 4 shows the accuracy of each model and Fig. 6 is the accuracy with changing the size of the training dataset.

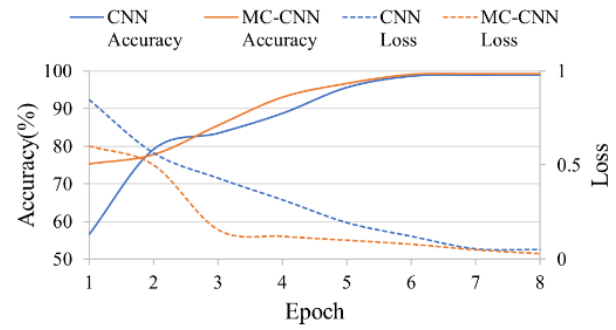
The experiment results illustrate two points. The first one is that our MC-CNN model performs the highest accuracy compared with conventional deep learning methods and machine learning methods. From CNN and MC-CNN, we find MC-CNN model has high stability and high accuracy contrast with other methods. multichannel allows CNN to make better use of information on every level. As shown in Fig. 4, MC-CNN has less loss and higher accuracy.

We try to detect other attacks by MC-CNN. There are 4 types of attacks in KDDCUP99. As shown in Table 5, in the face of 5 class tasks, MC-CNN also shows the good performance in each class. Though limited by the number of instances, the accuracy of U2R and R2L is higher than other models. It shows our DAD-MCNN can be used in other attack detection and ensure web service safety.

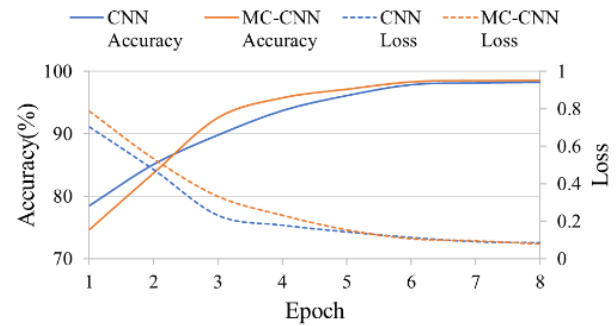
We change the size of the training dataset and evaluate the efficiency of CNN and MC-CNN. The second point is that MC-CNN is better in the limited dataset. We can see that at the 70% and 80% of original size, our model has higher accuracy. This characteristic helps us build DDoS detection system more easily when the training data is relatively insufficient.

Table 4. Accuracy of M-CNN and other models.

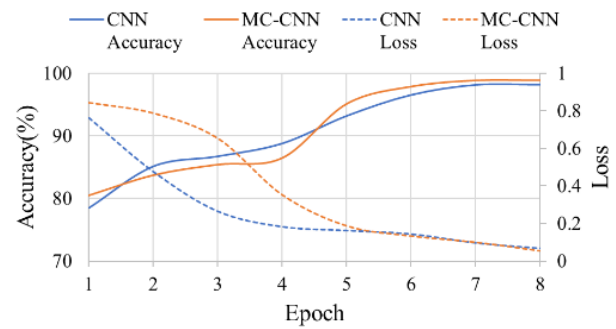
Dataset	MC-CNN	CNN	LSTM	RF	SVM	C4.5	KNN
KDDCUP99 (2-class)	99.18	98.87	96.53	98.97	98.13	97.17	98.97
KDDCUP99 (5-class)	98.54	98.17	95.60	98.35	97.98	96.83	98.13
CICIDS2017	98.87	98.11	97.86	95.63	95.46	92.60	93.54



(a) Accuracy and loss of KDDCUP99(2-class)



(b) Accuracy and loss of KDDCUP99(5-class)

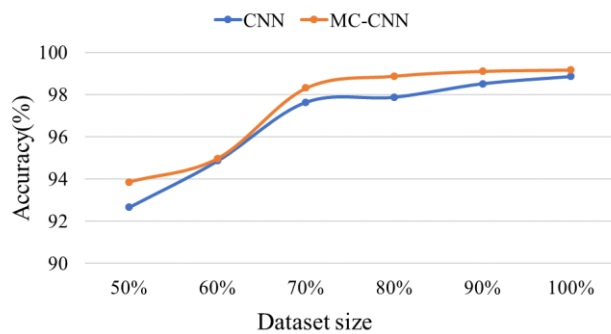


(c) Accuracy and loss of CICIDS2017

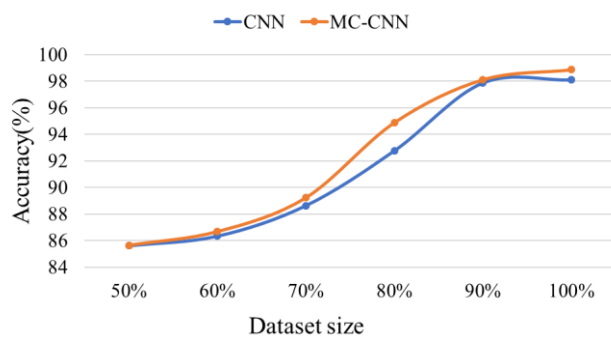
Figure 4. Accuracy and loss.

Table 5. Accuracy of 5-class in KDDCUP99.

Model	Normal	Dos	Probe	U2R	R2L
MC-CNN	99.32	99.18	89.63	35.16	16.53
CNN	99.03	98.92	85.71	30.74	12.95
LSTM	96.56	96.01	78.63	17.26	8.64
RF	98.78	98.42	75.03	3.52	2.63
SVM	98.19	98.02	72.63	10.41	5.79
C4.5	97.61	97.85	60.13	0	0
KNN	98.92	97.16	77.83	13.72	3.94



(a) KDDCUP99 (2-class)



(b) CICIDS2017

Figure 5. Different size of datasets.

5. CONCLUSION

In this paper, we propose the DAD-MCNN framework to detect DDoS attack via deep learning. In this framework, we propose a MC-CNN model that maximizes the use of feature information to make the outstanding detection. We perform experiments on KDDCUP99 and CICIDS2017. The results show that our framework has the highest accuracy contrasted with single channel CNN, LSTM and shallow models. When the training data is relatively insufficient, our framework also can make a good performance.

In the future, we will look for more attacks to test our MC-CNN model, improve the efficiency of detection framework and further optimize the performance of MC-CNN model.

6. ACKNOWLEDGMENTS

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